



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 2 (Summary and Tutorial)

Topic: Differentiation Techniques

Summary for Differentiation Techniques

1. DERIVATIVE AS A LIMIT

If $y = f(x)$, then $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$.

Eg: To prove that $\frac{d}{dx}(x^2) = 2x$ from 1st principles,

Let $y = f(x) = x^2$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^2 - x^2}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{x^2 + 2x\delta x + (\delta x)^2 - x^2}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} (2x + \delta x) \\ &= 2x\end{aligned}$$

2. BASIC RULES OF DIFFERENTIATION

Type of Rule	How it works	Examples
Distributive Rule	$\frac{d}{dx}[af(x) \pm bg(x)] = af'(x) \pm bg'(x)$, where a and b are constants.	
Product Rule	$\frac{d}{dx}[f(x)g(x)] = g(x)f'(x) + f(x)g'(x)$	$\begin{aligned}\frac{d}{dx}((2x-3)\sqrt{x-5}) \\ &= 2\sqrt{x-5} + (2x-3)\frac{1}{2\sqrt{x-5}} \\ &= \dots = \frac{6x-23}{2\sqrt{x-5}}\end{aligned}$
Quotient Rule	$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$	$\frac{d}{dx}\left(\frac{x+1}{x^2-3}\right)$

		$= \frac{(x^2 - 3)(1) - (x + 1)(2x)}{(x^2 - 3)^2}$ $= \dots = -\frac{x^2 + 2x + 3}{(x^2 - 3)^2}$
Chain Rule	1. $\frac{d}{dx} fg(x) = \frac{d}{dx} [f(g(x))] = [f'(g(x))] g'(x)$ or 2. $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ where $y = f(u)$ and $u = g(x)$	$\frac{d}{dx} (\sin \sqrt{x})$ $= \cos \sqrt{x} \times \frac{1}{2\sqrt{x}}$ $= \frac{1}{2\sqrt{x}} \cos \sqrt{x}.$
Relationship between $\frac{dx}{dy}$ and $\frac{dy}{dx}$	$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$	

Properties of logarithm (useful for simplifying expressions)

	Examples
a) $\ln(x^y) = y \ln x, x > 0$ b) $x^y = e^{\ln(x^y)}, x^y > 0$	$\frac{d}{dx} (2^{x+3}) = \frac{d}{dx} (e^{\ln 2^{x+3}})$ $= \frac{d}{dx} (e^{(x+3)\ln 2})$ $= e^{(x+3)\ln 2} \ln 2 = 2^{x+3} \ln 2$
c) $\log_x y = \frac{\log_z y}{\log_z x}, x, y, z > 0, x, z \neq 1$	$\frac{d}{dx} (\log_5 (x^2 + 1))$ $= \frac{d}{dx} \left(\frac{\ln(x^2 + 1)}{\ln 5} \right)$ $= \frac{1}{\ln 5} \times \frac{1}{x^2 + 1} \times 2x = \frac{2x}{(x^2 + 1)\ln 5}$

3. DERIVATIVES OF STANDARD FUNCTIONS

y	$\frac{dy}{dx}$
x^n for fixed n	nx^{n-1}
e^x	e^x
$\ln x$	$\frac{1}{x}$

y	$\frac{dy}{dx}$
$[f(x)]^n$	$n[f(x)]^{n-1} f'(x)$
$e^{f(x)}$	$e^{f(x)} f'(x)$
$\ln[f(x)], f(x) > 0$	$\frac{1}{f(x)} f'(x)$

Trigonometric Functions (Highlighted formulae are in MF27)

The following results hold for x measured in radians only.

If x is measured in degrees, then convert x° to $\frac{\pi x}{180}$ radians.

y	$\frac{dy}{dx}$	y	$\frac{dy}{dx}$
$\sin x$	$\cos x$	$\sin[f(x)]$	$[\cos(f(x))] f'(x)$
$\cos x$	$-\sin x$	$\cos[f(x)]$	$[-\sin(f(x))] f'(x)$
$\tan x$	$\sec^2 x$	$\tan[f(x)]$	$[\sec^2(f(x))] f'(x)$
$\cot x$	$-\operatorname{cosec}^2 x$	$\cot[f(x)]$	$[-\operatorname{cosec}^2(f(x))] f'(x)$
$\sec x$	$\sec x \tan x$	$\sec[f(x)]$	$[\sec(f(x)) \tan(f(x))] f'(x)$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$	$\operatorname{cosec}[f(x)]$	$[-\operatorname{cosec}(f(x)) \cot(f(x))] f'(x)$

Inverse Trigonometric Functions

The following results hold for x measured in radians only.

If x is measured in degrees, then convert x° to $\frac{\pi x}{180}$ radians.

y	$\frac{dy}{dx}$	y	$\frac{dy}{dx}$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}, \quad x < 1$	$\sin^{-1}[f(x)]$	$\frac{f'(x)}{\sqrt{1-[f(x)]^2}}, \quad f(x) < 1$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}, \quad x < 1$	$\cos^{-1}[f(x)]$	$-\frac{f'(x)}{\sqrt{1-[f(x)]^2}}, \quad f(x) < 1$
$\tan^{-1} x$	$\frac{1}{1+x^2}, \quad x \in \mathbb{R}$	$\tan^{-1}[f(x)]$	$\frac{f'(x)}{1+[f(x)]^2}, \quad f(x) \in \mathbb{R}$

4. HIGHER ORDER DERIVATIVES

In general, $\frac{d^n y}{dx^n} = f^{(n)}(x) = \frac{d}{dx} \left(\frac{d^{n-1} y}{dx^{n-1}} \right)$ for $n = 2, 3, 4, \dots$

Eg: If $y = x^n$, where n is a negative integer, then $\frac{dy}{dx} = nx^{n-1}$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (nx^{n-1}) = n(n-1)x^{n-2}$$

$$\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \dots = n(n-1)(n-2)x^{n-3}$$

5. IMPLICIT DIFFERENTIATION

For a relation involving x and y , for example $y^3 - x + y = 0$, which cannot be expressed in the form $y = f(x)$, we can obtain $\frac{dy}{dx}$ by differentiating every term in the equation with respect to x . Very often, Chain Rule will be applied.

Eg, to differentiate $y^3 - x + y = 0$ implicitly w.r.t. x ,

$$\frac{d}{dx}(y^3 - x + y) = 0$$

$$\Rightarrow \frac{d}{dx}(y^3) - \frac{d}{dx}(x) + \frac{d}{dx}(y) = 0$$

$$\text{Applying Chain Rule, } \frac{d}{dx}(y^3) = \frac{d}{dy}(y^3) \times \frac{dy}{dx} = 3y^2 \frac{dy}{dx}$$

$$\text{So we have } 3y^2 \frac{dy}{dx} - 1 + \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{1}{1+3y^2}$$

6. PARAMETRIC DIFFERENTIATION

Suppose $x = f(t)$ and $y = g(t)$, where $t \in \mathbb{R}$.

$$\text{By Chain rule : } \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \qquad \text{Recall } \frac{dt}{dx} = \frac{1}{\frac{dx}{dt}}.$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt}$$

Revision Tutorial Questions**Source of Question: ACJC JC1 CT1 9758/2017/Q2 modified**

- 1 Find $\frac{dy}{dx}$ in terms of x and y given that
- (i) $y = x^{\sin x}$, where $x > 0$, [3]
- (ii) $xy = \tan^{-1}(y^2)$. [2]

Source of Question: TJC JC1 CT1 9758/2017/Q7

- 2 (a) (i) Differentiate $\ln[\tan^{-1}(2x)]$ with respect to x . [2]
- (ii) Given that $y = \operatorname{cosec} x \cot x$, show that $\frac{dy}{dx} = -(\operatorname{cosec}^3 x + y \cot x)$. [2]
- (b) Given that $f(x) = \frac{x}{2} - 4 \ln x + \frac{15}{4} \ln(2x+1)$, show that $f'(x) = \frac{x^2 - 4}{x(2x+1)}$. [2]
- Hence show that the graph of $y = f(x)$ has exactly one stationary point. [2]
- Determine the nature of this stationary point without the use of a graphing calculator. [2]

Source of Question: DHS JC2 CT1 9758/2019/Q2**3 Do not use a graphing calculator in answering this question.**

- (i) Let $y = \frac{2x}{4+x^2} - \tan^{-1} \frac{x}{2}$. Show that $\frac{dy}{dx} = -\frac{4x^2}{(4+x^2)^2}$, and deduce that $y < 0$ for $x > 0$. [3]
- (ii) Hence show that if $f(x) = \frac{1}{x} \tan^{-1} \frac{x}{2}$, where $x > 0$, then f is a decreasing function. [2]

Source of Question: AJC JC1 CT1 9740/2016/Q3

4 (a) Given that $y^2 = 9 - \sin^2 x$, show that

(i) $2y \frac{dy}{dx} + \sin(2x) = 0$. [2]

(ii) $y \left(\frac{d^2 y}{dx^2} \right) + \left(\frac{dy}{dx} \right)^2 + 2y^2 = 17$. [4]

(b) Given that $|x| < \frac{1}{\sqrt{2}}$, find $\frac{d}{dx} \left[\sin^{-1} \left(2x\sqrt{1-x^2} \right) \right]$. Express your answer in the form $\frac{a}{\sqrt{1-x^2}}$, where a is a constant to be determined. [4]

Source of Question: EJC JC1 Promos 9758/2018/Q6

5 A curve D has parametric equations $x = a\theta - a \sin \theta$, $y = a - a \cos \theta$, where $a > 0$ and $0 < \theta < \pi$.

(i) Show that $\frac{dy}{dx} = \cot \frac{\theta}{2}$. [3]

(ii) Find the exact equation of the normal to the curve at the point for which $\theta = \frac{\pi}{3}$. [4]

Source of Question: MI Promos 9758/2018/Q12i

6 A curve C has parametric equations

$$x = 2 \cos 2t, \quad y = 3 \sin 2t, \quad \text{for } 0 \leq t \leq \frac{\pi}{4}.$$

Find $\frac{dy}{dx}$ in terms of t and hence show that as x increases, y decreases. [4]